# Pizza Restaurant Agentic System Report

**Github Repo Link:** https://github.com/Shehzaad-S/restaurant-agent-system-project

## Overview

This project implements a semi-autonomous multi-agent workflow designed to support restaurant inventory management, supplier planning, and food waste reduction. The system receives natural language sales reports from restaurant staff, converts them into structured sales data, updates inventory records, evaluates inventory health, and generates operational recommendations for restaurant managers. The primary objective of the system is to assist managers in making informed operational decisions while maintaining human oversight over all inventory and menu changes.

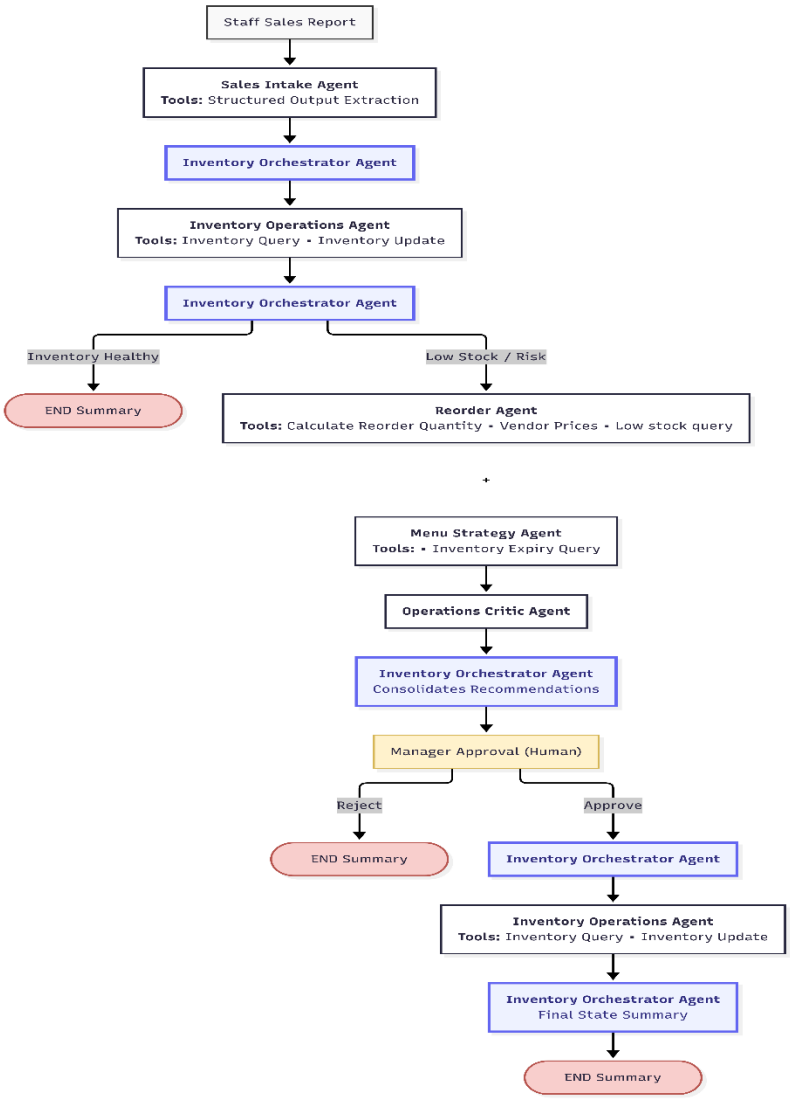
The workflow models a small Italian restaurant serving a limited menu of pizzas and pasta dishes. A specialized Sales Intake Agent transforms unstructured sales reports into validated business data. An Inventory Orchestrator Agent then coordinates workflow execution using a custom state machine and shared workflow state. Based on inventory conditions, the orchestrator routes work to specialized agents responsible for inventory analysis, supplier selection, menu optimization, and recommendation validation.

The system adopts a multi-agent architecture because the business problem consists of several distinct decision-making activities, including inventory assessment, procurement planning, menu strategy, and recommendation review. Separating these responsibilities into specialized agents improves modularity, transparency, and maintainability while allowing each agent to focus on a specific operational task.

Agents receive structured business information generated through predefined tool invocations and then apply LLM reasoning to generate recommendations, assessments, and critiques. This design improves reproducibility, transparency, and operational control while retaining the benefits of LLM-assisted reasoning.

Human approval is required before operational actions are executed, ensuring that AI-generated recommendations remain subject to managerial oversight. This approach aligns with recommendations that emphasize transparency, accountability, and human review when deploying systems that rely on probabilistic language models (Bender et al., 2021).

## System Architecture and Design




[Figure 1: Restaurant Agent System Architecture]

***Figure 1*** illustrates the architecture of the Restaurant Inventory Operations Agentic System. The workflow is coordinated by an Inventory Orchestrator Agent that manages workflow execution, maintains shared workflow state, and determines which specialized agents should be invoked at each stage of the process. The system uses a multi-agent architecture consisting of a Sales Intake Agent, Inventory Operations Agent, Reorder Agent, Menu Strategy Agent, and Operations Critic Agent.

The workflow begins when a restaurant employee submits a natural language sales report. The Sales Intake Agent transforms the report into structured sales data using Gemini Flash

and Pydantic model validation. This structured output is passed to the Inventory Orchestrator Agent, which initiates workflow execution through a custom state machine.

The Inventory Operations Agent is responsible for inventory-related calculations, including ingredient consumption analysis, inventory updates, and inventory health assessment. This agent interacts directly with SQLite-backed business data, including recipes, inventory records, and supplier information. Based on the resulting inventory assessment, the orchestrator determines whether inventory conditions are healthy or whether corrective actions are required.

When inventory risks are detected, the workflow invokes two specialized recommendation agents. The Reorder Agent evaluates low-stock ingredients, supplier pricing, and delivery times to generate replenishment recommendations. The Menu Strategy Agent evaluates expiry-risk ingredients and proposes menu specials designed to reduce food waste and improve inventory utilization. These agents work independently but operate on the same shared workflow state.

Recommendations are subsequently reviewed by the Operations Critic Agent, which serves as a reflection and validation layer. The critic evaluates inventory feasibility, recommendation quality, and operational alignment before producing an approval assessment, confidence score, and supporting feedback. The resulting recommendations are then presented to a human manager for final approval.

A significant design decision was the use of deterministic tool invocation combined with LLM-based reasoning. While the project incorporates reasoning-and-action concepts similar to those described by ReAct (Yao et al., 2023), the agents do not dynamically select tools during execution. Instead, business tools are invoked through predefined workflow steps to retrieve inventory, recipe, pricing, and expiry information. Agents then reason over this structured information to generate recommendations and decisions. This approach improves transparency, reproducibility, and operational control while still benefiting from LLM-assisted reasoning.

The system maintains workflow context through a shared WorkflowState object rather than conversational memory. Session memory was intentionally excluded because the application operates as a workflow system rather than a multi-turn conversational agent. The WorkflowState object stores sales reports, inventory assessments, recommendations, critic reviews, manager decisions, and workflow metrics. This shared state design allows agents to remain loosely coupled and supports independent agent testing by providing only the state required for a specific task.

Persistent business memory is provided through SQLite, which stores inventory records, recipes, supplier pricing, and menu information. SQLite was selected because the workflow operates primarily on structured operational data rather than unstructured knowledge.

Finally, a custom state-machine architecture was implemented to manage workflow routing, approval handling, retry behavior, and workflow termination. This design makes workflow transitions explicit, improves observability, and simplifies testing and evaluation. During testing, the state machine successfully demonstrated conditional routing, human-in-the-loop approval, and retry behavior when recommendations were rejected before final approval.

## Decision Logic and Behaviour

The system follows a state-driven decision-making process in which each agent receives structured information from the shared WorkflowState and contributes a specialized decision. Overall we tried to solidify the system to become dependable in a business setting, once success is found we can test additional reasoning capabilities.

The Sales Intake Agent is responsible for transforming natural language manager reports into structured sales records. During testing, valid reports were successfully converted into structured sales data, enabling downstream inventory analysis. However, when intentionally ambiguous sales reports were provided, the agent returned an empty SalesReport, demonstrating that recommendation quality depends on the completeness and clarity of manager-provided information.

The Reorder Agent combines inventory assessments with supplier pricing and delivery information to identify replenishment needs. Rather than selecting the lowest-cost supplier in all cases, the agent reasons about both cost and operational risk. For example, during testing the agent selected Fresh Foods Wholesale for low-stock ingredients because its shorter delivery time reduced the likelihood of inventory shortages, despite a slightly higher price per kilogram.

The Menu Strategy Agent evaluates ingredients approaching expiry and generates menu specials designed to reduce waste while remaining operationally practical. During execution the agent consistently prioritized ingredients with the shortest remaining shelf life and produced recommendations that incorporated multiple expiry-risk ingredients into a single menu item.

The Operations Critic Agent acts as a reflection layer that reviews reorder and menu recommendations before they are presented for approval. The agent evaluates

recommendation quality, operational feasibility, and alignment with business objectives before providing an approval decision, confidence score, and supporting rationale. This additional validation step increases transparency and provides managers with an explanation of why recommendations were generated.

The workflow also responds differently depending on operational conditions. When inventory is healthy, the workflow may terminate after analysis. When inventory risks are detected, recommendation agents are invoked and recommendations are reviewed by both the Critic Agent and a human manager. If recommendations are rejected, the state machine routes the workflow through a retry path before a final decision is reached. This behavior was successfully demonstrated during testing and illustrates the workflow's ability to adapt to different operational situations while maintaining human oversight.

## Safety, Reliability and Transparency

Several safeguards were incorporated to improve the safety, reliability, and transparency of the workflow. First, all agent outputs are validated using Pydantic models, ensuring that information exchanged between agents conforms to predefined schemas before it enters the workflow. This reduces the likelihood of malformed or unexpected agent outputs propagating through the system.

Second, the workflow uses deterministic tool invocation rather than unrestricted autonomous tool selection. Inventory queries, recipe lookups, vendor comparisons, and database updates are executed through predefined workflow steps, making agent behavior more predictable and easier to audit. Agents reason over structured business data rather than dynamically selecting tools at runtime.

Third, an Operations Critic Agent provides a reflection and validation layer before recommendations reach management. This additional review step increases transparency by generating approval decisions, confidence scores, and supporting justifications for each recommendation.

The most important safeguard is the human-in-the-loop approval process. Inventory replenishment and menu changes are not executed automatically. A manager must explicitly approve recommendations before operational updates are applied, ensuring that final decision-making authority remains with a human operator. This design aligns with recommendations emphasizing transparency, accountability, and appropriate human oversight when deploying systems that rely on probabilistic language models (Bender et al., 2021).

Finally, the workflow uses centralized logging, token tracking, and workflow metrics to improve observability. These mechanisms make workflow decisions, agent activity, execution costs, and retry behavior visible to system operators, supporting both evaluation and future auditing efforts.

## Observed Behaviour and Limitations

Evaluation testing demonstrated that the workflow consistently executed the intended inventory management process from sales intake through recommendation generation, review, approval, and execution. Workflow logs showed that agents successfully exchanged information through the shared WorkflowState, allowing inventory assessments, reorder recommendations, menu strategies, and critic reviews to be coordinated without direct dependencies between agents.

One notable observation was the effectiveness of the human-in-the-loop approval stage. During testing, manager approval was intentionally rejected, causing the state machine to enter a retry path and regenerate recommendations before returning for a second approval review. The workflow subsequently completed successfully, demonstrating that the orchestration layer could adapt to changing workflow conditions while maintaining managerial oversight.

A key limitation was identified during invalid input testing. When an intentionally ambiguous manager report was provided, the Sales Intake Agent returned an empty SalesReport rather than generating inaccurate data. While this behavior prevented hallucinated sales records, the workflow continued using the existing inventory state and generated recommendations despite no sales activity being recorded. This revealed a weakness in workflow entry-point validation and suggests that future versions should require at least one valid sales record before inventory analysis begins.

Testing also highlighted a trade-off between oversight and efficiency. When recommendations were rejected and regenerated, token usage increased from approximately 5,900 tokens to more than 11,000 tokens, nearly doubling execution cost and runtime. This behavior improved review quality and control but increased workflow resource consumption.

Finally, repeated executions occasionally generated different menu special names and recommendation wording despite identical inventory conditions. Although the recommendations remained operationally valid and aligned with business objectives, this

behavior reflects the probabilistic nature of LLM-based systems and demonstrates that recommendation phrasing may vary between executions.

**Ethical and Responsible Use Considerations**

A primary ethical consideration in this project is accountability for operational decisions generated by AI-assisted workflows. While the system can recommend inventory purchases, menu changes, and supplier selections, the underlying language models remain probabilistic systems that may occasionally generate incomplete or suboptimal recommendations. As noted by Bender et al. (2021), language model outputs may appear authoritative despite lacking genuine understanding of the domain in which they operate.

This risk is particularly relevant because approved recommendations directly modify business records and may influence purchasing decisions. During testing, the workflow demonstrated that valid-looking recommendations could still be generated even when an empty Sales Report was provided. Although the recommendations remained operationally reasonable, this behavior highlights the importance of validating input data and maintaining human oversight before recommendations are executed.

A second consideration is automation bias, where users may place excessive trust in AI-generated recommendations because they appear well-reasoned and professionally presented. The workflow therefore exposes recommendation justifications, confidence scores, and critic feedback to improve transparency and encourage managerial review rather than blind acceptance of AI-generated outputs.

Overall, the system is intended to function as a decision-support tool rather than an autonomous decision maker. Human review remains essential to ensure accountability, data quality, and responsible operational decision making.

**Future Improvements**

Several practical enhancements could improve the robustness, safety, and analytical capabilities of the system. At the workflow level, future versions should validate sales reports before execution begins and require corrected input when insufficient sales information is detected. Additional safeguards could prevent recommendation generation when critical workflow inputs are missing.

From an agent-design perspective, the current workflow intentionally favors deterministic tool usage combined with LLM reasoning. Future iterations could introduce a dedicated data analysis agent that adopts a more ReAct-style approach (Yao et al., 2023), allowing the agent to dynamically investigate inventory trends, supplier information, or operational anomalies before generating recommendations. Similarly, a Retrieval-Augmented

Generation (RAG) component could be incorporated to provide access to supplier documentation, operational policies, historical purchasing records, and business knowledge sources.

Additional enhancements include historical sales forecasting, advanced profitability analysis, and workflow analytics. The existing database schema already includes a workflow_runs table that could be expanded to support execution history, approval tracking, token consumption analysis, and operational performance reporting. Together, these improvements would increase the system's decision making capabilities while preserving the transparency and human oversight established in the current design.

## References


Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the dangers of stochastic parrots: Can language models be too big?* Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21), 610–623. https://doi.org/10.1145/3442188.3445922

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